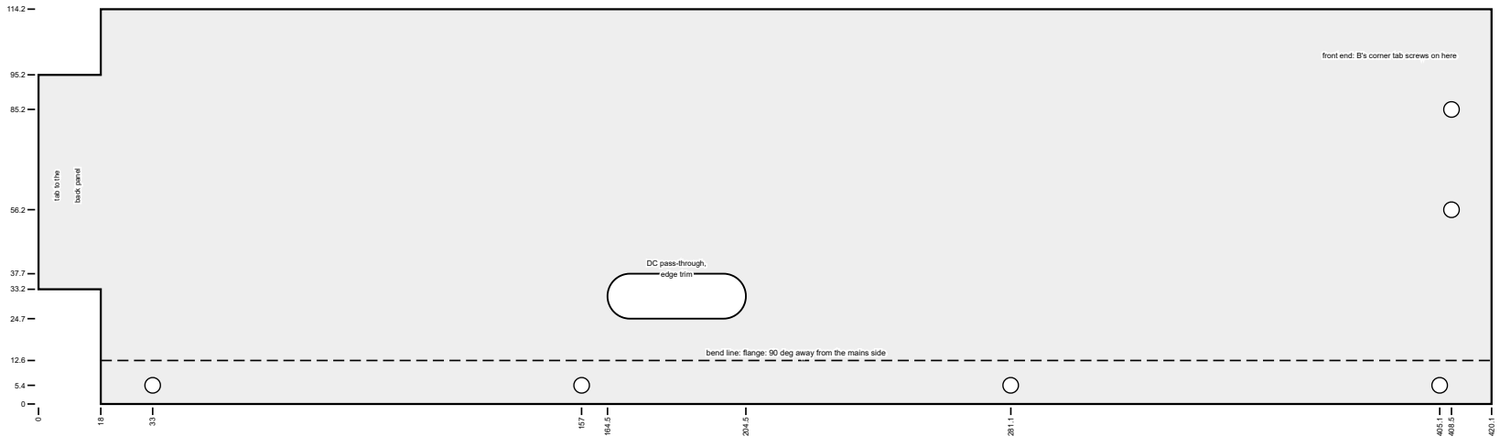
RMR control panel: mains compartment wall

Two pieces of 1/16 in 5052 that close the 110 VAC compartment on the lower deck (sheet 3, view B), cut from the offcut of the upper plate's sheet. Flat patterns and section 1:2, plan 1:5, printed at 100 %. Millimetres.

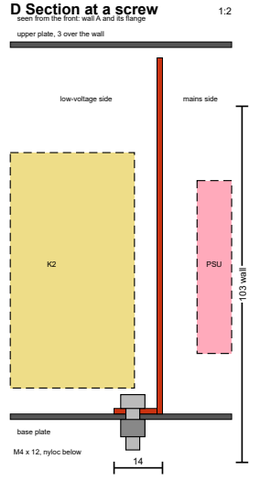
A Wall A, flat, seen from the mains side

420.1 x 114.2; along x, mains face at y 182



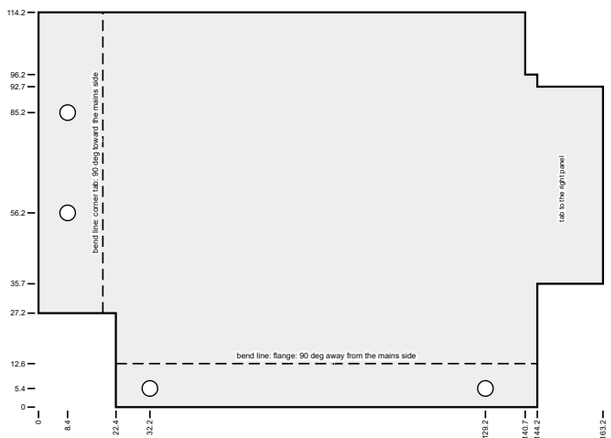
D Section at a screw

1:2



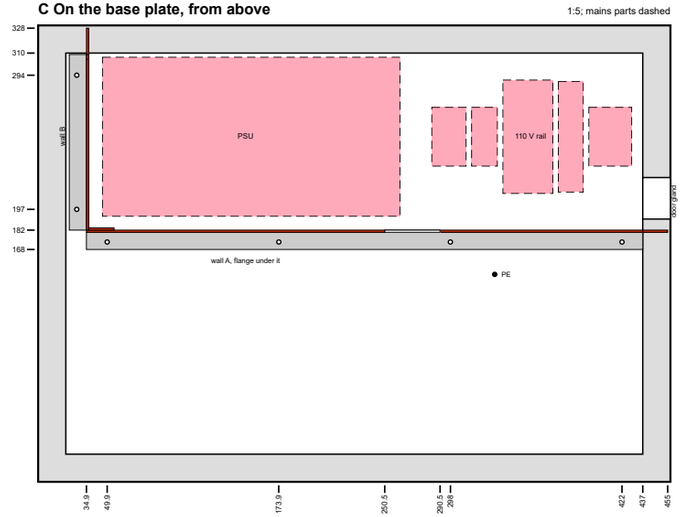
B Wall B, flat, mains side

163.2 x 114.2; along y



C On the base plate, from above

1:5; mains parts dashed



Material and cut

1/16 in 5052-H32 from the offcut of aluminum 2, beside the upper plate (sheet 4): flat A 420.1 x 114.2, B 163.2 x 114.2. Shear the blanks, nibble the end tabs and B's corner relief, drill the Ø4.5 holes, cut the pass-through, deburr; DXF: wall_mains_flat.dxf (layers CUT and BEND).

Bends

90 deg on a brake, inside radius one thickness. K 0.42 is assumed: bend deduction 2.8 per bend. Bend a test strip first and correct WallK if its legs come out long or short. Flanges fold away from the mains side; B's corner tab folds toward it and lies on A's mains face.

Fastening and PE

Flanges: 6 M4 x 12 into the base plate, nyloc below, 3.8 mm past the nut (NutFits). Corner: 2 M4 x 12 through B's tab and A, heads on the mains side. Drill the base plate through the flange holes in place. Each piece is its own PE part: a ring lug under the flange screw at x 49.9 on A and y 197 on B, with no part over it, since the lug lifts the head.

Openings

End tabs reach between the rings to 2 mm from the back and right panels. B's tab, and a notch in the top of B's right end, keep 1 mm from the front right corner brackets (BracketChecks). The top stops 3 mm under the upper plate. DC pass-through 40 x 13, 20 over the deck, centred on the mains duct: PSU V+ and V- out. Move it to the PSU's terminal end once that face is known. The 110 V door cable leaves by an M20 gland in the back panel (u 205), not through the wall.

Upper plate

Drill no DIN rail or duct screw within 4.8 mm of the wall's top edge: A at y 180.4 to 182, B at x 34.9 to 36.5. Its nyloc and tail hang 9.4 mm; the wall stops 3 under the plate.

Fit

Before the lower deck's parts: flange screws, then B's corner tab to A, then the PE jumpers. Check that the upper module drops on clear.

Mains compartment wall

Sheet 5. Legal, landscape, 100 %.
Flat patterns and section 1:2, plan 1:5.
Designed 2026-10-18 (issue #146).
DXF: wall_mains_flat.dxf.